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Determine la resistencia nominal de la viga detallada en la figura. Considere hormigón G18 y acero A630-420H. Datos: $r := 5\ cm$ (al centroide de la fila inferior)

Sección 35x70

2 barras del 10

5 barras del 22

Datos

$b := 35\ cm \qquad h := 70\ cm$

$n_b := 5 \qquad \phi_b := 22\ mm \qquad r := 5\ cm \qquad d := h - r = 65\ cm$

$$A_s := \pi \cdot \frac{\phi_b^{\,2}}{4} \cdot n_b = 19.0066\ cm^2$$

$n'_b := 2 \qquad \phi'_b := 10\ mm \qquad d' := r = 5\ cm$

$$A'_s := \pi \cdot \frac{\phi'^{\,2}_b}{4} \cdot n'_b = 1.5708\ cm^2$$

$f_c := 180\ \frac{kgf}{cm^2} \qquad f_y := 4200\ \frac{kgf}{cm^2} \qquad \beta := 0.85$

$n_{filas} := 1 \qquad \phi_s := 10\ mm$

$r_{est} := 3\ cm$ (al borde libre del estribo)

$$s := \frac{b - 2 \cdot r_{est} - \frac{n_b}{n_{filas}} \cdot \phi_b - 2 \cdot \phi_s}{\frac{n_b}{n_{filas}} - 1} = 4\ cm$$

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Solución

(Suponer que A'_s y A_s fluyen)

VIGA 1

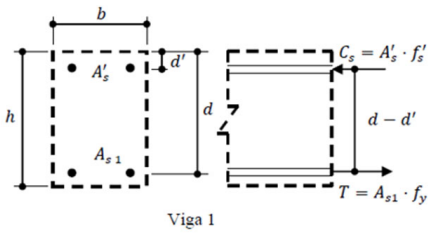
Equilibrio de fuerzas

T = C

$A_{s1} \cdot f_y = A'_s \cdot f_y$

$A_{s1} := A'_s = 1.5708\ \text{cm}^2$

$M_{n1} := A_{s1} \cdot f_y \cdot (d - d') = 3.9584\ \text{tonnef m}$



Viga 1

VIGA 2

$A_{s2} := A_s - A_{s1} = 17.4358\ \text{cm}^2$

Equilibrio de fuerzas

C = T

$0.85 \cdot f_c \cdot b \cdot a := A_{s2} \cdot f_y$

$a := \dfrac{A_{s2} \cdot f_y}{0.85 \cdot f_c \cdot b} = 13.6752\ \text{cm}$

$c := \dfrac{a}{\beta} = 16.0884\ \text{cm}$

$M_{n2} := A_{s2} \cdot f_y \cdot \left(d - \dfrac{a}{2}\right) = 42.5926\ \text{tonnef m}$

$M_n := M_{n1} + M_{n2} = 46.551\ \text{tonnef m}$

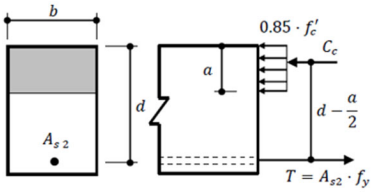
$\varepsilon_c := 0.003$

$\varepsilon_s := \dfrac{\varepsilon_c}{c} \cdot (d - c) = 0.0091$

$\varepsilon_s > \varepsilon_y$ Fila de aceros en tracción ==> Se verifica el supuesto

$\varepsilon'_s := \dfrac{\varepsilon_c}{c} \cdot (c - d') = 0.0021$

$\varepsilon'_s > \varepsilon_y$ Fila de aceros en compresión ==> Se verifica el supuesto



Viga 2

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